

Data Warehousing and Data Mining

CSIT — 7th Semester — 2080 — Answer Sheet
Subject Code: CSC410

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. State Apriori property. Find frequent itemsets and association rules from the transaction database given below using Apriori algorithm. Assume min. support is 50% and min confidence is 75%. Transaction ID Items Purchased
1 Juice 2 Juice 3 Bread, Butter 4 Juice, Milk 5 Cheese, Juice, Milk

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. How classification differs from regression. Train ID3 classifier using the dataset given below. Then predict class label for the data
Age Competition Type Profit (Class Label)
old Yes SW Down old No SW Down old No HW Down
mid Yes SW Down mid Yes HW Down mid No HW Up
mid No SW Up new Yes SW Up new No HW Up
new No SW Up

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Why the concept of data mart is important? Discuss different data warehouse scheme with examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. How OLAP differ from data mining? Explain various stages of KDD with suitable block diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Discuss different ways of smoothing noisy data along with suitable examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. How many cuboids are possible from 5-dimensional data? Discuss the concept of full cube and iceberg cube.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. How K-medoids clustering differs from K-means clustering? Divide the following data points into two clusters using k-medoids algorithm. Show computation upto 2 iterations. {(70,85), (65,80), (72,88), (75,90), (60,50), (64,55), (62,52), (63,58)}

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Discuss working of DBSCAN algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Which algorithm is used for training multi-layer perceptron? Discuss the algorithm in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain the OLAP operations with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Discuss the concept of multimedia data mining along with the concept of similarity search.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write down short notes on a. Support Vector Machine b. Multi-dimensional Data Model

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.